SVEUČILIŠTE U ZAGREBU
FAKULTET ORGANIZACIJE I INFORMATIKE
VARAŽDIN

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IGRA INTERAKTIVNE FIKCIJE - INFORM 7

PROJEKT

DEKLARATIVNO PROGRAMIRANJE

Varaždin, 2025.

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Izjava o izvornosti

Izjavljujem da je ovaj projekt izvorni rezultat mojeg rada te da se u izradi istoga nisam koristio drugim izvorima osim onima koji su u njemu navedeni. Za izradu rada su korištene etički prikladne i prihvatljive metode i tehnike rada.

Autorica potvrdila prihvaćanjem odredbi u sustavu FOI Radovi

Sažetak

Ovaj rad bavi se korištenjem deklarativnog programiranja u jeziku Inform 7 kroz izradu detektivske igre pod nazivom "A Nutter Mystery". U uvodnom dijelu objašnjava se motivacija za odabir ove teme, naglašavajući kako Inform 7 nudi pristup programiranju putem prirodnog jezika, čineći proces intuitivnijim i kreativnijim za autore interaktivnih priča. Teorijski dio rada istražuje osnovne karakteristike deklarativnog programiranja te ključne strukture Informa 7: sobe, predmete, osobe, pravila i relacije. Svaka od tih struktura omogućuje detaljno modeliranje interaktivnih svjetova i narativa, a njihova implementacija objašnjava se u kontekstu kreiranja detektivske priče. Kritički osvrt analizira prednosti i nedostatke ovog pristupa. Premda je rad s Informom 7 bio zanimljiv i kreativan, suočila sam se s određenim ograničenjima jezika, poput osjetljivosti na redoslijed riječi i izazova u stvaranju suptilnih tragova koji nisu odmah očiti igraču. Na kraju sam prikazala implementaciju same igre koja uključuje dizajn prostorija, pronalazak skrivenih tragova i dijaloge s osumnjičenima. Zaključno, projekt je spojio programiranje i kreativno pisanje, pružajući jedinstveno iskustvo razvoja interaktivne fikcije.

Ključne riječi: Inform 7, Interaktivna fikcija, detektivska igra, pravila, priča, tragovi

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1. Uvod

Od malena su me fascinirali kriminalistički romani - njihova sposobnost da nas uvuku u svijet misterija, intriga i logičkog povezivanja tragova nikad mi nije dosadila. Zato sam odlučila napraviti vlastitu igru koja kombinira elemente krimića i humora, pružajući igračima priliku da riješe neobičnu misteriju dok uživaju u zabavnom tonu priče.

Za realizaciju projekta odabrala sam Inform 7; programski jezik koji omogućuje stvaranje interaktivnih fikcija koristeći prirodni jezik. Oduševila me njegova filozofija "programiranja riječima" jer, kao netko tko uživa u pisanju i izražavanju kroz tekst, smatram da takav pristup pruža kreativniju i intuitivniju alternativu uobičajenim programskim jezicima.

Inform 7 dizajniran je za razvoj tekstualnih avantura, u kojima igrači istražuju virtualne svjetove, manipuliraju objektima i komuniciraju s likovima kroz tekstualne naredbe. Jezik je deklarativne prirode, što znači da se fokusira na definiranje pravila i relacije u igri, dok sam sustav brine o njihovom provođenju. Ova pristupačnost, zajedno s mogućnošću da se programiranje približi pisanju proze, činila ga je idealnim izborom za realizaciju mog projekta.

Nakon pomnog planiranja priče, likova i tragova, nastala je igra zvana "A nutter mystery" koju ću detaljnije objasniti kasnije u dokumentu.

2. Razrada teme

Deklarativno programiranje je paradigma programiranja koja se fokusira na opisivanje što program treba učiniti, za razliku od imperativnih jezika koji zahtijevaju detaljan opis kako nešto postići. Važno je istaknuti kako u deklarativnom programiranju programer definira pravila i željeni rezultat, dok se implementacijski detalji prepuštaju programskom jeziku ili sustavu na kojem se program izvršava. [1] Neki poznatiji deklarativni jezici su [2]: SKJUEL, HTML, Prolog, SML, Ocaml, Haskell, Erlang i naravno, Inform 7, koji je bio korišten za ovaj projekt.

U svojoj srži, Inform 7 koristi predikate i pravila da bi modelirao odnose i ponašanje u virtualnom svijetu. Jezik omogućuje autorima definiranje svijeta, objekata i njihovih međusobnih odnosa kroz rečenice koje su u skladu s engleskom sintaksom. Formalizam koji podržava Inform 7 temelji se na predikatnom računu prvog reda. Svaka rečenica u programu predstavlja implicitni predikat ili pravilo.

2.1. Inform 7

Interaktivna fikcija, ponekad nazvana tekstualna avantura ili IF, oblik je računalne književnosti u kojem čitatelj (ili "interaktor") upravlja radnjama glavnog lika unosom običnih rečenica na tipkovnici. IF je započeo krajem 1970-ih, imao komercijalni vrhunac tijekom 1980-ih, a posljednjih godina doživljava značajan povratak popularnosti [3]. Sustavi za stvaranje interaktivne fikcije dostupni su javnosti od 1987., ali Inform 7, predstavljen 2006., prvi je alat koji autorima omogućuje pisanje izvornog teksta koristeći obične engleske rečenice.[4]

Inform 7 je programski jezik dizajniran za kreiranje interaktivne fikcije koji je poznat po svojoj sintaksi koja koristi prirodan jezik. Ovaj pristup omogućuje programerima da pišu kod koji je čitljiv i intuitivan, približavajući proces programiranja pisanju proze. Može se koristiti za stvaranje prototipa igrica, kao pomoć u pisanju književnih djela te čak i za edukativne svrhe. [5]

Kreator određuje na koji način igrač može stupiti u interakciju sa elementima igre. Svaka struktura ima definirana pravila opisa i korištenja u kodu, a za dodatne elemente se mogu koristiti ekstenzije. Osnovne strukture u Informu 7 su sobe, predmeti, osobe, pravila i relacije. Ukratko ću objasniti svaku od ovih struktura.

2.1.1. Sobe

Sobe (engl. *rooms*) čine osnovu igre i predstavljaju različite lokacije koje igrač može istraživati. Svaka soba može imati opis te stvari i osobe koje se nalaze u njoj, a međusobno su povezane relacijama koje određuju kako igrač prelazi između njih (npr. sjever, jug, gore, dolje i sl.). Korisno je u opisu sobe napisati u kojim se smjerovima iz te sobe smije kretati igrač. Primjer opisa sobe iz moje igrice je na slici 1.

```
The Library is a room. "A sanctuary of knowledge turned suspicious. The heavy scent of old books and polished wood fills the air, while rows of towering shelves cast long
  shadows. A grand desk sits in the corner, its surface cluttered with papers that seem oddly out of place, as if someone had been searching for something. Most of the books
  are dusty, it's obvious they haven't been touched for years. But a couple of them were obviously recently taken from the dusty shelves.

From here you can go south to the Living Room."
```

Slika 1: Primjer kreiranja sobe u Inform 7 igri; vlastita izrada

2.1.2. Predmeti

Predmeti (engl.*things*) su elementi unutar soba s kojima igrač može stupiti u interakciju, poput ključeva, knjiga, hrane ili kutija. Svaki predmet može imati opis i svojstva poput težine, boje, vidljivosti, jestivosti itd. Stvari se također mogu definirati kao kutije (eng.*container*), što znači da mogu biti zaključane, otvorene kao i sadržavati druge stvari u sebi koje igrač inicijalno ne vidi. Interakcija sa stvarima je zanimljiva jer se dodatnim pravilima stvarima mogu dodati zanimljivi elementi. Primjerice, uz pravu ekstenziju može se omogučiti igraču interakcija s računalom na način da može pretraživati razne upite. Igrač stvari može uzeti, pogledati, otvoriti, pojesti itd. U mojoj igri stvari služe za pronalaženje tragova i prikupljanje informacija. Na slici 2 vidi se primjer stvari koja ima obilježje "fixed in place", što znači da ga igrač ne može uzeti.

```
A fireplace is in the Living room. It is fixed in place. "The fireplace is cold and filled with a layer of ash, the charred remains of logs scattered within. Something faintly catches
  the eye—barely noticeable amid the soot, as if hiding in plain sight."
```

Slika 2: Primjer kreiranja stvari u Inform 7 igri; vlastita izrada

2.1.3. Osobe

Osobe (engl. *persons*) predstavljaju likove unutar igre, bilo da su saveznici, neprijatelji ili neutralni promatrači. Oni mogu odgovarati na igračeve interakcije i pružati informacije, zadatke ili prepreke. U mojoj igri svi su likovi osumnjičeni za ubojstvo osim jednog koji igra ulogu detektiva. Svakog je lika moguće pitati za određene stvari pod uvjetom da su pronađene. Primjer stvaranja osobe nalazi se na slici 3.

```
SpongeBob is a suspect. He is in the staff room. The description is "SpongeBob is, as always, full of energy, but there's a strange heaviness to his usual wide-eyed optimism
  tonight. His square, yellow sponge body and big blue eyes are instantly recognizable. His youthful grin masks a deeper tension beneath the surface. His movements are a
  little twitchy, and he seems to be avoiding certain suspects in the room, though you can't quite tell why."
```

Slika 3: Primjer kreiranja osobe u Inform 7 igri; vlastita izrada

2.1.4. Pravila

Pravila (engl. *rules*) definiraju ponašanje sustava te kako igra reagira na igračeve akcije. Pravila mogu biti opća (primjenjiva uvijek) ili specifična za određene situacije. Na slici 4 prva rečenica je primjer općeg pravila koje onemogućuje prikaz igraču je li stvar kutija ili ne kad uđe u sobu, a ispod toga je specifično pravilo koje određuje što će se dogoditi kad igrač otvori jednu specifičnu stvar.

```
Rule for printing room description details of a closed container: stop.

After opening the suit of armor:
    say "You open the suit of armor with a creak. Inside, you find a hidden key.";
    move the S key to player;
    now the S key is found;
```

Slika 4: Primjer kreiranja pravila u Inform 7 igri; vlastita izrada

2.1.5. Relacije

Relacije (engl. *relations*) su logički odnosi među strukturama, poput "igrač se nalazi u sobi" ili "ključ otključava vrata". One pomažu u modeliranju međusobne povezanosti elemenata igre. Primjerice, na slici 5 prikazana je relacija između dva predmeta (ključa i škrinje) te igrača.

```
Instead of unlocking the rose chest with the rose key:
    say "You turn the rose key in the lock, and with a satisfying click, the chest opens. Inside, you find a letter from Simon's dad.";
    now the letter is found;
    now the rose chest is open.
```

Slika 5: Primjer kreiranja relacije u Inform 7 igri; vlastita izrada

2.2. Kritički osvrt

Rad na projektu u Informu 7 bio je jedno od najzanimljivijih iskustava tijekom studija, prvenstveno zbog toga što se programiranje u ovom jeziku oslanja na prirodni jezik, a ne na klasičnu programersku logiku. Kao netko tko voli pisanje i izražavanje kroz riječi, Inform 7 mi je pružio priliku da programiram na način koji se osjeća više kao pričanje priče nego pisanje koda.

Unatoč tome što mi je rad s Informom 7 bio izrazito zanimljiv, tijekom projekta sam se susrela s određenim ograničenjima jezika. Primjerice, poredak riječi u sintaksi ima ključnu ulogu, pa su se ponekad naredbe koje su izgledale ispravno odbijale izvršiti samo zbog drugačijeg redoslijeda riječi. To je često izazivalo frustracije, jer bih nakon istraživanja i konzultiranja dokumentacije otkrila da je rješenje puno jednostavnije nego što se na prvi pogled činilo.

Osim tehničkih izazova, projekt je donio i koncepcijske dileme. Kao ljubiteljica krimića, željela sam kreirati misteriju s elementima suptilnosti, gdje igrač postepeno shvaća priču i dolazi do završnog zaključka kroz vlastita istraživanja. Međutim, priroda interaktivne fikcije otežava ovu suptilnost jer Inform 7 automatski prikazuje sve relevantne informacije čim igrač uđe u prostoriju. To je rezultiralo time da su tragovi u igri često bili preočiti.

Pokušala sam nadoknaditi taj nedostatak tako što sam omogućila igračima slobodu u pristupu: nisu bili prisiljeni ispitivati likove ili pratiti sve tragove, već su sami mogli odlučiti koje radnje žele poduzeti. Ipak, svjesna sam da bi bolja implementacija uključivala sofisticiraniji sustav tragova koji bi bili manje očiti, omogućujući igračima dublje razmišljanje i vlastito donošenje

zaključaka. Zato sam napravila nekoliko završetaka priče koji otkrivaju različite priče koje sve povezuju. Time ova igra odskače od tipičnog kriminalističkog žanra jer kraj u biti ne otkriva pravog ubojicu.

Unatoč izazovima, uživala sam u kreativnom procesu oblikovanja svijeta igre, likova i priče. Svaka poteškoća, poput neočekivanih grešaka u kodu ili potreba za prilagodbom koncepta, naučila me važnosti fleksibilnosti i upornosti u razvoju projekata. Na kraju, Inform 7 mi je omogućio da spojim svoju ljubav prema pisanju i programiranju u jednom projektu, a sama igra rezultat je tog procesa.

Sve u svemu, ovo iskustvo ostavilo je pozitivan dojam na mene. Otkrila sam novi žanr igrica te uspješno napravila vlastitu kojom se ponosim. Možda ću ju u budućnosti još nadograditi novim elementima i dodatno obogatiti priču.

3. Implementacija

3.1. Komponente

Prostori (engl. *Rooms*)

Glavni prostori uključuju dnevnu sobu, glavni ulaz, veliku dvoranu, Simonovu sobu, vrt, kuhinju, knjižnicu, mali hodnik, blagavaonicu i sobu od posluge. Svaka soba ima jedinstven opis i povezana je s drugim prostorima, čineći bogatu mrežu lokacija za istraživanje unutar vile u kojoj se događa radnja.

Posebna vrsta sobe je tzv. mračna soba (engl.*dark room*) u kojoj igrač ne vidi ništa (ne dobije opis sobe, kao ni nabrojane stvari koje se nalaze u sobi) te ne može stupiti u interakciju s predmetima koji se nalaze u toj sobi dok ne uzme svjetiljku. Primjer uvjeta za osvjetljavanje mračne sobe nalazi se na slici 6.

```
Before looking in the Garden:
    if the Flashlight is not carried by the player:
        continue the action;  [Room stays dark.]
    otherwise:
        now the Garden is lit;
        continue the action.

After looking in the Garden when the Flashlight is carried:
    now the Garden is dark;
```

Slika 6: Primjer osvjetljavanja mračne sobe u Inform 7 igri; vlastita izrada

Osumnjičeni (engl. *Suspects*)

Likovi poput Shreka, Super Maria, SpongeBoba, Smurfette i Shakire nalaze se u igri kao potencijalni osumnjičeni. Svaki lik ima svoj detaljan opis i ponašanje prilikom ispitivanja. Na osumnjičenima se može provesti početno ispitivanje (prije no što su ikakvi tragovi nađeni) te za svaki trag koji upućuje na njih. Oni povlače pozadinske priče kroz igru. Igrač u bilo kojem trenutku može završiti igru upisivanjem naredbe "guess" uz ime osumjičenog kojeg optužuju za ubojstvo. Duck-tective je jedini NPC koji ne spada u osumnjičene. On vodi igrača sarkastičnim komentarima i savjetima. Njegova prisutnost dodaje humor igri, a igrač ga može pitati za pomoć oko neutralnih elemenata poput ključeva i kovčega. Primjer odgovora Duck-tectiva na igračeva pitanja nalazi se na slici 7.

```
Instead of asking Duck-tective about "rose chest":
    say "Duck-tective: [italic type]So you need a hint about where the key might be? The ROSE key that opens the ROSE chest? Get out
      of my sight apprentice! [roman type]";

Instead of asking Duck-tective about "remote":
    say "Duck-tective: [italic type]Oh, a remote? There's certainly a reason it's there.[roman type] But don't waste your time looking for a
      TV, alright? This isn't some sitcom where you get to sit back and relax. Quack![roman type]";

Instead of asking Duck-tective about "yellow note":
    say "Duck-tective: [italic type]A yellow note with a single word? Oh, how mysterious![roman type] You don't need me to spell it out
      for you. Or do you?[roman type] He squints, waiting for a response. [italic type]QUACK!! It's a password, you donut! Get out of
      here while you still have an ounce of dignity left![roman type]";
Instead of asking Duck-tective about "computer":
    if the computer is not found:
        say "Duck-tective: [italic type]Oh, a locked computer? How *original*.[roman type] You don't need me to tell you that it needs
          a password, do you? Or maybe you're just going to keep staring at it until it magically unlocks? Come on, genius, figure it
          out—there's a reason you're standing here. Don't make me do all the thinking for you![roman type]";
    otherwise:
        say "Duck-tective: [italic type]So you found some interesting stuff on the computer, eh? How about you go interrogate our
          lovely Colombian singer now? I'm curious if she's shaking her hips in oblivion, or if she knew about all this. Maybe she's got
          more than just those famous moves to keep under wraps![roman type] ";
```

Slika 7: Primjer razgovora s likom u igri; vlastita izrada

Predmeti (engl. *Things*)

Ključni predmeti uključuju projektnu dokumentaciju, ružin ključ, fotografiju, pismo i još mnogo toga. Ovi predmeti često otkrivaju tragove ili omogućuju napredak u igri. Tragovi imaju element "found" koji obilježava nađene tragove, što je bitno za razgovor s osumnjičenima. Također, određeni tragovi se nalaze u virtualnoj prostoriji "Hidden storage" (kojoj igrač nema pristup) dok ih igrač ne otkrije gledanjem ispod ili iza vidlljivih elemenata, kako ih ne bi vidio pri ulasku u sobu. Na slici 8 vidi se primjer istih.

```
The burnt letter is in the Hidden storage. The burnt letter can be found. The burnt letter is not found. The description is "The charred
  remains of a letter lie buried in the ashes, its text mostly obliterated by fire. Only a few fragmented words remain legible: 'tonight', 'my
  love', 'end', 'the armor' and 'Simon's money'. Look's like someone has big plans for tonight.

On the back of the burnt letter, just barely spared from the flames, are faint marks resembling a doodle—a tiny heart encircling the word
  'Smurfabulous'. It's the kind of smurfy flourish only one person would ever use."

A fireplace is in the Living room.  It is fixed in place. "The fireplace is cold and filled with a layer of ash, the charred remains of logs
  scattered within. Something faintly catches the eye—barely noticeable amid the soot, as if hiding in plain sight."

Instead of examining the fireplace:
    if the burnt letter is not found:
        say "In the ashes you find the pieces of a burnt letter.";
        Move the burnt letter to the Living room;
        now the burnt letter is found;
    otherwise:
        say "You've already checked inside the fireplace.";
```

Slika 8: Primjer skrivenih tragova u Inform 7 igri; vlastita izrada

Posebne Radnje (engl. *Custom Actions*)

[6] Radnje poput `interrogate`, `look under`, i `read` omogućuju igraču posebnu definiranu interakciju s likovima i predmetima. `Interrogate` se može koristiti za ispitivanje osumnjičenih kako bi se saznale dodatne informacije o određenim predmetima. `Look under` je radnja koja se primjenjuje na predmetima kako bi igrač mogao pogledati ispod svakog predmeta. Slično funkcionira radnja `look behind`. `Read` je radnja koja omogućuje igraču da pročita napisani sadržaj nekog predmeta koji se može čitati (primjerice knjige ili pisma). Na slikama 9 i 10 prikazana je implementacija naredba `interrogate` i `read`.

```
Interrogating is an action applying to one thing. Understand "interrogate [someone]" as interrogating.
A suspect can be interrogated or uninterrogated. A suspect is usually uninterrogated.

Check interrogating:
    if the noun is not a suspect:
        say "That's not someone you can interrogate." instead.

Shakira-interrogated is a truth state that varies. Shakira-interrogated is false.
Shrek-interrogated is a truth state that varies. Shrek-interrogated is false.
Mario-interrogated is a truth state that varies. Mario-interrogated is false.
Smurfette-interrogated is a truth state that varies. Smurfette-interrogated is false.
SpongeBob-interrogated is a truth state that varies. SpongeBob-interrogated is false.

Carry out interrogating:
    if the noun is uninterrogated:
        now the noun is interrogated;
        if the noun is Shakira:
            now Shakira-interrogated is true;
            say "Shakira: [italic type]Ah, cariño, I work in the same industry as Simon, and let me tell you, it's not uncommon to make
              enemies with some very powerful people. But Dios mío, I never expected this! We used to be close, working together a
              lot, but not so much lately. Simon and I even did a collab on a song once, back when he was still part of a group. Ay,
              perdón, I'm still in shock... ¡qué locura! Who in their right mind could have done such a horrible thing?[roman type]";
```

Slika 9: Implementacija radnje "interrogate" u Inform 7 igri; vlastita izrada

```
A thing has some text called printing. The printing of a thing is usually "blank".
Understand the command "read" as something new. Understand "read [something]" as reading. Reading is an action applying to one
  thing, requiring light. Check reading: if the printing of the noun is "blank", say "Nothing is written on [the noun]." instead. Carry out
  reading: say "You read: [printing of the noun][line break]". Report reading: do nothing.

The printing of the family history book is "You open the book to find a family tree inside. It shows Simon's lineage, with one name
  scratched off under 'Father'. The name is unreadable, and the picture is not shown, but this detail seems significant. Ahh and of course
  next to Simon's picture there were his brothers, Alvin and Theodore, from whom he was unseparable in his younger days. May their little
  souls rest in piece."
```

Slika 10: Implementacija radnje "read" u Inform 7 igri; vlastita izrada

Mapa

Karta omogućuje igraču bolje razumijevanje lokacija u igri. Dizajnirana je da prikaže raspored prostorija kako bi igrač, nakon što ju uzme, mogao bilo kad pogledati u kartu i lakše se orijentirati po prostorijama. Primjer mape nalazi se na slici 11.

```
A map is in the Grand Hall.
The description of the map is "[map art]".

To say map art:
    say "                                    N[line break]";
    say "                                  [line break]";
    say "                        Simon's room       Library[line break]";
    say "                                |                  |[line break]";
    say "                                |                  |[line break]";
    say "                          Small Hall ------ Living Room ---- Terrace[line break]";
    say " W                             |                                          E [line break]";
    say "                                |                  [line break]";
    say "              Garden ----Grand Hall --- Dining Room ------ Kitchen[line break]";
    say "                                |                 |[line break]";
    say "                                |                 |[line break]";
    say "                        Main Entrance    Staff Room[line break]";
    say "                                [line break]";
    say "                                    S";
```

Slika 11: Stvaranje mape u Inform 7 igri; vlastita izrada

Tablica ispitivanja

Napravila sam tablicu mogućih odgovora ispitanika s obzirom na to koji su tragovi do tog trenutka nađeni u igrici. Za to mi je trebao atribut `found` za tragove. Način na koji sam to

implementirala vidljiv je na slikama 12. i 13.

```
otherwise:
    let responses-found be 0;
    say "You begin questioning [the noun].";
    repeat through the Table of Suspect Responses:
        if the suspect entry is the noun and the clue entry is found:
            say "[response entry][paragraph break]";
            increase responses-found by 1;
    if responses-found is 0:
        say "[The noun] says, 'I already told you, I have nothing to do with this!'[paragraph break]";
```

Slika 12: Stvaranje tablice dijaloga u Inform 7 igri (1/2); vlastita izrada

```
Table of Suspect Responses
Suspect     Clue                Response
Mario       the photo           "Mario: [italic type]Si, I-a saw the photo of my mamma and that chipmunk-a. Simon said-a they were old friends-a. Why is that
  important now amici?[roman type]"
Mario       the letter          "Mario: [italic type]I-a didn't know about-a any letter! Mamma mia! You're telling me Simon and I were family? How did I-a not
  know this-a? Sorry, I'm-a little confused. Does that mean I'm no longer a suspect? I would never hurt my own family! You gotta believe me, amico![roman type]"
Mario       the mushroom        "Mario: [italic type]Oh, the mushroom-a? That's just-a my lucky snack! I must-a dropped it while I was-a running around. I-a never
  leave home without one, you know. It's-a not suspicious, amici![roman type]"
Smurfette   the burnt letter    "Smurfette: [italic type]Oh my smurf! You think I would write a letter about smurfing my boyfriend? That's absolutely smurfing
  ridiculous! I may be a little dramatic, but I'm not that crazy. I would never smurf something like that![roman type]"
Smurfette   the hidden chest    "Smurfette: [italic type]Well... of course I knew where it was. I was his girlfriend, for smurf's sake! But that doesn't prove anything!
  Simon had amazing qualities other than being rich. Like... uh... he was good at singing! And... um... he had great taste in hats![roman type]"
SpongeBob   the sticky note     "SpongeBob: [italic type]What? A threatening note? Oh, barnacles! That wasn't supposed to be threatening—it was just... stern
  encouragement! I was just trying to, uh, motivate the chef to make better krabby patties... I mean, food! Yeah, that's it![roman type]"
SpongeBob   the spatula         "SpongeBob: [italic type]Oh, that old thing? Uh... I must've dropped it after making some, uh, really rare... steaks! Yeah, that's it!
  Steaks! It's just grease, I swear! Besides, a spatula's a cook's best friend—I'd never use it for anything bad![roman type]"
SpongeBob   the burger          "SpongeBob: [italic type]Ah, so you've tried the old Krabby Patty! Glad to have a new costumer! Sorry, I can't tell you the recipe.
  It's a secret. [roman type]"
Shakira     the sheet music     "Shakira: [italic type]Si, I know about Chaka Chaka. Everyone knows it. They play it on the radio more than the original! Just like all
  of his copycat songs... Acorn you, Smells Like Chipmunk Spirit... why does every single one of his tracks have to be a chipmunk pun?[roman type]"
Shakira     the poster          "Shakira: [italic type]Well, that's... an interesting message. I always encourage people to dance to the rhythm of their own beat, but
  I don't think I quite meant it that way. But hey, he was a fan, right? I actually find it kinda funny and ironic—guess I'm just flattered someone's so into my vibe!
  [roman type]"
Shakira     the financial paper "Shakira: [italic type]I'm not surprised. There are always those who try to ride on the coattails of others, hoping for a quick rise to
  fame. Once I noticed what he was doing I was so mad. I guess no amount of money can buy originality.[roman type]"
Shakira     the computer        "Shakira: [italic type]¿Qué? The messages on the computer? No, I haven't seen them. But if what you're saying is true, there's
  definitely enough evidence to finally arrest that thief manager! He's the real criminal here![roman type]"
Shrek       the document        "Shrek: [italic type]Aye, I did see that document. And they say I'm the monster! Honestly, that wee squirrel's lucky I didn't gobble
  him up right there and then.[roman type]"
Shrek       the wooden bat      "Shrek: [italic type]A murder weapon? Ye've got to be kiddin' me! I could crush that wee squirrel with me bare hands if I wanted
  to! Let's just say that bat had... well... other purposes.[roman type]"
Shrek       the mirror          "Shrek: [italic type]Aw, come on, that was just a wee joke! Just tryin' to scare the little rat. He came to mess with the wrong ogre.
  [roman type]"
```

Slika 13: Stvaranje tablice dijaloga u Inform 7 igri (2/2); vlastita izrada

3.2. Prikaz igre

U dolje navedenim slikama bit će prikazani neki od dijelova igre u kojima se otkrivaju pojedini tragovi te interakcije s okolinom, stvarima i osobama.

The living room of the Victorian mansion buzzed with chatter, the clinking of glasses, and the enchanting voice of a singer filling the air. After all, a New Year's Eve party like this was a rare occasion.

When the clock struck midnight, fireworks illuminated the sky outside, and the guests erupted into cheers of "Happy New Year!" The sound of toasts and laughter echoed throughout the room.

Moments later, the lights flickered back on, and an eerie silence replaced the joyful music. Gasps and murmurs spread as all eyes fell on the lifeless body of Simon the chipmunk sprawled on the carpet, blood pooling around him.

Duck-tective burst dramatically into the room, his detective hat slightly askew. You followed close behind.

"It appears," he quacked with flair, "we have a MURDER on our hands!"

He pointed a feathery wing at the scene. "Apprentice! This is your moment. Search the mansion, gather evidence, interrogate the suspects, and do whatever it takes to crack this case. This is your final test to earn your badge as a true detective!"

With a sly grin, Duck-tective added, "Of course, I already know who the culprit is. If you find yourself lost, you may consult me... but only if you wish me to question your capabilities. Now go forth and question the suspects-quack, quack!"

****INFORMATION FOR THE PLAYER****

You can guess the murderer when you find all the clues. If you know to which character the clue is tied to, you can ask that character about it. For neutral clues like keys and chests, ask Duck-tective for help.

Try looking inside or behind the items you see in every room.

If you want to try guessing the killer before finding all the clues type "guess (suspect)" - this will, however, end the game

Slika 14: Početak igre; vlastita izrada

A Nutter Mystery
An Interactive Fiction by Laura Kriskovic
Release 1 / Serial number 250116 / Inform 7 build 6M62 (I6/v6.33 lib 6/12N) SD

Staff Room
A small, functional room where the staff can catch their breath. The furniture is sparse-just a couple of old chairs and a cluttered table with papers scattered across it. The air feels thick with unease as all the suspects have been brought here until the investigation is over. They sit in silence, each one lost in their thoughts, glancing nervously at each other. After you gather up the initial information, you can ask them about the clues you find to get information about it.

Go ahead and interrogate the suspects!

From here you can go north to the Dining Room.

You can see Shrek the ogre, Super Mario, SpongeBob, Smurfette and Shakira here.

>interrogate shrek
Shrek: *Well, to be honest, I didn't really like tha^ guy. Don't hold it against me, but he was a right rich snob, he was! Aye, I dinnae think he deserved to be killed, but och, he sure knew how t^make a person want t^kill him! Hah! What, did I say somethin^ wrong?*
>interrogate shakira
Shakira: *Ah, cariño, I work in the same industry as Simon, and let me tell you, it's not uncommon to make enemies with some very powerful people. But Dios mío, I never expected this! We used to be close, working together a lot, but not so much lately. Simon and I even did a collab on a song once, back when he was still part of a group. Ay, perdón, I'm still in shock... ¡qué locura! Who in their right mind could have done such a horrible thing?*
>n

dining room
The dining room exudes grandeur, with a long, polished oak table set for a feast that never came. Ornate chairs, upholstered in rich velvet, are carefully arranged around it, though one seems hastily pushed back, as if someone stood up abruptly. Crystal chandeliers hang from the ceiling, their glow reflecting off fine china and silverware.

From here you can go east towards the Kitchen, south to the Staff Rooms or west to get to the Grand Hall.

You can see a burger here.

>

Slika 15: Početno ispitivanje osumnjičenih; vlastita izrada

Grand Hall
The centerpiece of the mansion, the grand hall exudes luxury with its gleaming marble floor and an opulent chandelier that sparkles above. Yet, the wide staircase and empty corridors seem to echo with whispers, as if the room itself is watching.

From here you can go west out the back door to the Garden, south towards the Main Entrance, east to the Dining Room or north up stairs towards the Small Hall.

You can see a map, a wooden bat, a flashlight and a suit of armor here.

>take map
Taken.

>look at map

```
                                N

          Simon's room        Library
                |                |
                |                |
            Small Hall ------ Living Room ---- Terrace
W               |                                          E
                |
     Garden ----Grand Hall --- Dining Room ------ Kitchen
                |                |
                |                |
          Main Entrance     Staff Room

                                S
```

>w

Darkness
It is pitch dark, and you can't see a thing.

>

Slika 16: Prikaz mape i mračne sobe; vlastita izrada

Darkness
It is pitch dark, and you can't see a thing.

>e

Grand Hall
The centerpiece of the mansion, the grand hall exudes luxury with its gleaming marble floor and an opulent chandelier that sparkles above. Yet, the wide staircase and empty corridors seem to echo with whispers, as if the room itself is watching.

From here you can go west out the back door to the Garden, south towards the Main Entrance, east to the Dining Room or north up stairs towards the Small Hall.

You can see a wooden bat, a flashlight and a suit of armor here.

>take flashlight
Taken.

>w

Garden
The garden is pristine, every bush and flower meticulously arranged to perfection, as if prepared for a royal visit. The scent of roses and freshly mowed grass lingers in the crisp night air. A majestic fountain sparkles at the center, but nearby, a once-proud marble sculpture lies shattered on the ground, its pieces scattered like a puzzle. The damage looks fresh, and the footprints leading away suggest someone was here-and left in a hurry. From here you can go east to go to the Grand Hall.

From here you can go east to go to the Grand Hall.

You can see a rose bush and a statue here.

>

Slika 17: Osvjetljavanje mračne sobe svjetiljkom; vlastita izrada

Living room
This is the room where it happened. Though the police have removed the body, the faint metallic scent of blood still lingers in the air. The grand furniture and ornate decor now feel cold and uninviting, as if the room itself remembers the crime.

From here you can go north to the Library, west to the Small Hall or east to get to the Terrace.

The fireplace is cold and filled with a layer of ash, the charred remains of logs scattered within. Something faintly catches the eye-barely noticeable amid the soot, as if hiding in plain sight.

You can also see Duck-tective, a mushroom, a spatula, a piano (on which is a sheet music), a stage and a couch here.

>e

terrace
The terrace offers a breathtaking view of the garden below, though the cold night air seems to carry a sense of unease. Elegant wrought-iron furniture is neatly arranged, yet a single chair is overturned, hinting at a moment of panic or a hurried escape.

From here you can go west to the Living room.

You can see a vase of roses here.

>examine vase
A beautiful vase of roses stands on a little stool.

>look under vase
You find a key hidden under the vase.

>examine key
A small key with a rose engraving. It looks like it might fit something special.

>

Slika 18: Gledanje ispod objekata i skriveni objekti; vlastita izrada

library
A sanctuary of knowledge turned suspicious. The heavy scent of old books and polished wood fills the air, while rows of towering shelves cast long shadows. A grand desk sits in the corner, its surface cluttered with papers that seem oddly out of place, as if someone had been searching for something. Most of the books are dusty, it's obvious they haven't been touched for years. But a couple of them were obviously recently taken from the dusty shelves.

From here you can go south to the Living Room.

You can see a family history book, a document and a folder here.

>look at book
An old, worn book titled "Family Ties and Legacy". It looks like it might contain something important.

>read book
You read: You open the book to find a family tree inside. It shows Simon's lineage, with one name scratched off under "Father". The name is unreadable, and the picture is not shown, but this detail seems significant. Ahh and of course next to Simon's picture there were his brothers, Alvin and Theodore, from whom he was unseparable in his younger days. May their little souls rest in piece.

>

Slika 19: Čitanje kao radnja; vlastita izrada

Simon's room
Simon's room feels more like a miniature world than a traditional bedroom. Tiny, ornate furniture is arranged with meticulous precision, from a walnut-shell bed lined with silk to a thimble-sized lamp on the nightstand. The air is sweet with the faint scent of acorns, and an open chest in the corner overflows with shiny trinkets and treasures. Despite its charm, a few scattered papers on the floor and a knocked-over stool suggest that not everything was as perfectly in place as it seemed.

From here you can only go south to the Small Hall.

You can see a bed, a mirror, a poster and a computer here.

>examine bed
It's pretty big for a chipmunk...almost the size of a cat's bed. Is there something hidden under it?

>look under bed
You find a photograph here.

>examine photograph
You can't see any such thing.

>examine photo
A photograph of a chipmunk standing next to a female gnome eating a pizza. The writing on the back says, "From Rose to her little brother."

>

Slika 20: Pronalaženje skrivenih objekata; vlastita izrada

Garden
Garden
The garden is pristine, every bush and flower meticulously arranged to perfection, as if prepared for a royal visit. The scent of roses and freshly mowed grass lingers in the crisp night air. A majestic fountain sparkles at the center, but nearby, a once-proud marble sculpture lies shattered on the ground, its pieces scattered like a puzzle. The damage looks fresh, and the footprints leading away suggest someone was here-and left in a hurry. From here you can go east to go to the Grand Hall.

From here you can go east to go to the Grand Hall.

You can see a rose bush and a statue here.

>look at bush
A bush with vibrant roses. Looks like something shines behind it.

>look behind bush
There's a small chest there.

>unlock chest with rose key
You turn the rose key in the lock, and with a satisfying click, the chest opens. Inside, you find a letter from Simon's dad.

>read letter
You read: A letter from his father's will. It says:

To my beloved son Simon,

If you are reading this, it means I am no longer with you. I am sorry for the distance between us, both physical and emotional, over the years. Life has a way of stealing precious time, and I deeply regret the moments we lost.

Slika 21: Otvaranje škrinja odgovarajućim ključem; vlastita izrada

Staff Room

From here you can go east towards the Kitchen, south to the Staff Rooms or west to get to the Grand Hall.

You can see a burger here.

>look at burger
Someone took only one bite of this Krabby Patty. I wonder if it was poisoned or simply bad.

>eat burger
You stuff the burger in your face. You finally understand the hype about Krabby Patties. You won't even care if they're poisoned.

>s

Staff Room
A small, functional room where the staff can catch their breath. The furniture is sparse-just a couple of old chairs and a cluttered table with papers scattered across it. The air feels thick with unease as all the suspects have been brought here until the investigation is over. They sit in silence, each one lost in their thoughts, glancing nervously at each other. After you gather up the initial information, you can ask them about the clues you find to get information about it.

Go ahead and interrogate the suspects!

From here you can go north to the Dining Room.

You can see Shrek the ogre, Super Mario, SpongeBob, Smurfette and Shakira here.

>interrogate spongebob
SpongeBob: *Haha! Chipmunk?? What chipmunk? I have NEVER seen one in my life! OH SIMON THE CHIPMUNK! Of course I know Simon! He was always hanging around the Krusty Krab... He even employed me to make Krabby Patties for some parties he was hosting. Why do you ask?..... NO, I'M NOT HIDING ANYTHING!! YOU'LL NEVER CATCH ME ALIVE!* He runs away, slipping on some butter.

>interrogate spongebob
You begin questioning SpongeBob.
SpongeBob: *Ah, so you've tried the old Krabby Patty! Glad to have a new costumer! Sorry, I can't tell you the recipe. It's a secret.*

Slika 22: Ispitivanje osumnjičenog nakon pronalaska traga; vlastita izrada

>guess shrek
You accuse Shrek, and Duck-tective rolls his eyes.

"Shrek found out Mario was behind a project destroying his beloved swamp, so he set off to avenge his home. Armed with a wooden bat, he arrived at Simon's house, ready to wreak havoc. He started by smashing a statue in the garden when Simon opened the door. Realizing the trouble he was in, Simon invited Shrek to the party, trying to calm him down. Shrek, though furious, eased up a little after a few Krabby Patties."

"But things took a turn when Shrek snooped around Simon's house and found a document confirming the project was already set in motion, claiming minimal environmental impact. Enraged by the deceit, Shrek couldn't hold back. At the party, in front of everyone, he stomped down hard on Simon, crushing him completely. No one even noticed as his tiny body was crushed underfoot." "Shrek just stood there, panting. His face twisted in frustration. "I couldn't let him get away with it. He was ruin" everything... ma family... ma swamp," Shrek muttered to himself, rubbin" his brow. "But now... now it's safe. Too bad it had to end like this."

Duck-tective lets out a long sigh. "A sad way for things to end...but that is the tragic way of life."

*** The End ***

Would you like to RESTART, RESTORE a saved game, QUIT or UNDO the last command?
>

Slika 23: Optuživanje osumnjičenog i završetak igre; vlastita izrada

4. Zaključak

Igra kombinira bogatu priču, raznovrsne lokacije i interaktivne elemente kako bi stvorila jedinstveno iskustvo za igrače. Posebna pažnja posvećena je razvoju likova, tako da svaki od njih ima jedinstvenu osobnost i doprinos radnji. Osim toga, svjetlosni elementi i skriveni tragovi dodaju dodatni sloj strategije i istraživanja. Dodavanjem dinamičkih radnji poput ispitivanja osumnjičenih te upotrebom predmeta za otključavanje novih dijelova priče, igra omogućuje igraču da osjeti stvarnu kontrolu nad radnjom. Sam kraj igre omogućuje igraču da na neki način odabere završetak priče bez njegovog inicijalnog saznanja.

Jedinstvena komponenta igre je i humor koji donose karakteri NPC-eva, balansirajući ozbiljnost misterije i čineći igru zabavnom i privlačnom za širu publiku. Sve to čini ovu igru ne samo zanimljivom, već i iskustvom koje potiče na ponovnu igru kako bi se otkrili svi detalji i alternativni završetci.

Razvoj igre u Informu 7 omogućio je intuitivan i strukturiran način implementacije svih ključnih elemenata, uključujući složene odnose između likova, predmeta i prostorija. Inform 7 se pokazao kao izvanredan alat za dizajn interaktivne fikcije, omogućujući lako povezivanje narativnih elemenata i stvaranje prilagođenih radnji. Njegova fleksibilnost i prirodan jezik znatno su olakšali razvoj i testiranje, dok su ugrađeni alati omogućili detaljno testiranje i poboljšanje iskustva igrača.

Zaključno, igra je osmišljena s ciljem stvaranja interaktivnog i uzbudljivog iskustva koje kombinira elemente istraživanja, strategije i narativa, dok humor i razrađeni likovi unose posebnu toplinu i osobnost u cijelu avanturu. Kroz bogatstvo detalja i mogućnosti, igra nudi izazov i zadovoljstvo za svakog igrača, a korištenje Inform 7 značajno je doprinijelo realizaciji ove vizije.

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5. Prilog

Projekt se nalazi na linku: `https://github.com/lkriskovi21/projekti/tree/main/projekt`